brane into and through the bulk. The black holes terminate in sharp points called “singularities.” The wormholes connect one region of our brane to another. As usual, I suppress one of our brane’s three dimensions, so the brane looks like a two-dimensional surface.



Fig. 4.5. Black holes and wormholes extending out of our brane into and through the bulk. One space dimension is removed from both our brane and the bulk. *[Drawing by the artist Lia Halloran.]*

Tidal Gravity

Einstein’s relativistic laws dictate that planets, stars, and unpowered spacecraft near a black hole move along the straightest paths permitted by the hole’s warped space and time. Figure 4.6 shows examples of four such paths. The two purple paths headed into the black hole begin parallel to each other. As each path tries to remain straight, the two paths get driven toward each other. The warping of space and time drives them together. The green paths, traveling circumferentially